MSG No. 129.412 $P4/nmm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: -4m21'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[1,6,7,8,10,11,12,13,17,22,23,24,26,27,28,29]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,3,4,5,9,14,15,16,18,19,20,21,25,30,31,32]

Table 2: Wyckoff site: 2b, site symmetry: -4m21'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,6,7,8,10,11,12,13,17,22,23,24,26,27,28,29]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,3,4,5,9,14,15,16,18,19,20,21,25,30,31,32]

Table 3: Wyckoff site: 2c, site symmetry: 4mm1'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1,2,3,6,12,13,15,16,17,18,19,22,28,29,31,32]
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[4,5,7,8,9,10,11,14,20,21,23,24,25,26,27,30]

Table 4: Wyckoff site: 4d, site symmetry: ..2/m1'

No.	position	mapping
1	$[0, 0, 0]$	[1,8,9,16,17,24,25,32]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,4,10,12,18,20,26,28]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,5,11,13,19,21,27,29]
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,7,14,15,22,23,30,31]

Table 5: Wyckoff site: 4e, site symmetry: ..2/m1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,8,9,16,17,24,25,32]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,4,10,12,18,20,26,28]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[3,5,11,13,19,21,27,29]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[6,7,14,15,22,23,30,31]

Table 6: Wyckoff site: 4f, site symmetry: 2mm.1'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[1,6,12,13,17,22,28,29]
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[2,3,15,16,18,19,31,32]
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[4,5,9,14,20,21,25,30]
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	[7,8,10,11,23,24,26,27]

Table 7: Wyckoff site: `8g`, site symmetry: `..21'`

No.	position	mapping
1	$[x, -x, 0]$	`[1,8,17,24]`
2	$[x+\frac{1}{2}, x, 0]$	`[2,4,18,20]`
3	$[-x, \frac{1}{2}-x, 0]$	`[3,5,19,21]`
4	$[\frac{1}{2}-x, x+\frac{1}{2}, 0]$	`[6,7,22,23]`
5	$[-x, x, 0]$	`[9,16,25,32]`
6	$[\frac{1}{2}-x, -x, 0]$	`[10,12,26,28]`
7	$[x, x+\frac{1}{2}, 0]$	`[11,13,27,29]`
8	$[x+\frac{1}{2}, \frac{1}{2}-x, 0]$	`[14,15,30,31]`

Table 8: Wyckoff site: `8h`, site symmetry: `..21'`

No.	position	mapping
1	$[x, -x, \frac{1}{2}]$	`[1,8,17,24]`
2	$[x+\frac{1}{2}, x, \frac{1}{2}]$	`[2,4,18,20]`
3	$[-x, \frac{1}{2}-x, \frac{1}{2}]$	`[3,5,19,21]`
4	$[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}]$	`[6,7,22,23]`
5	$[-x, x, \frac{1}{2}]$	`[9,16,25,32]`
6	$[\frac{1}{2}-x, -x, \frac{1}{2}]$	`[10,12,26,28]`
7	$[x, x+\frac{1}{2}, \frac{1}{2}]$	`[11,13,27,29]`
8	$[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}]$	`[14,15,30,31]`

Table 9: Wyckoff site: `8i`, site symmetry: `.m.1'`

No.	position	mapping
1	$[\frac{1}{4}, y, z]$	`[1,12,17,28]`
2	$[\frac{1}{2}-y, \frac{1}{4}, z]$	`[2,15,18,31]`
3	$[y, \frac{1}{4}, z]$	`[3,16,19,32]`
4	$[\frac{3}{4}, -y, -z]$	`[4,9,20,25]`
5	$[\frac{3}{4}, y+\frac{1}{2}, -z]$	`[5,14,21,30]`
6	$[\frac{1}{4}, \frac{1}{2}-y, z]$	`[6,13,22,29]`
7	$[y+\frac{1}{2}, \frac{3}{4}, -z]$	`[7,10,23,26]`
8	$[-y, \frac{3}{4}, -z]$	`[8,11,24,27]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m1'`

No.	position	mapping
1	$[x, x, z]$	`[1,16,17,32]`
2	$[\frac{1}{2}-x, x, z]$	`[2,12,18,28]`
3	$[x, \frac{1}{2}-x, z]$	`[3,13,19,29]`
4	$[x+\frac{1}{2}, -x, -z]$	`[4,10,20,26]`
5	$[-x, x+\frac{1}{2}, -z]$	`[5,11,21,27]`
6	$[\frac{1}{2}-x, \frac{1}{2}-x, z]$	`[6,15,22,31]`
7	$[x+\frac{1}{2}, x+\frac{1}{2}, -z]$	`[7,14,23,30]`

continued ...

Table 10

No.	position	mapping
8	$[-x,\, -x,\, -z]$	`[8,9,24,25]`

Table 11: Wyckoff site: `16k`, site symmetry: `11'`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1,17]`
2	$\left[\frac{1}{2}-y,\, x,\, z\right]$	`[2,18]`
3	$\left[y,\, \frac{1}{2}-x,\, z\right]$	`[3,19]`
4	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	`[4,20]`
5	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	`[5,21]`
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[6,22]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[7,23]`
8	$[-y,\, -x,\, -z]$	`[8,24]`
9	$[-x,\, -y,\, -z]$	`[9,25]`
10	$\left[y+\frac{1}{2},\, -x,\, -z\right]$	`[10,26]`
11	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\, y,\, z\right]$	`[12,28]`
13	$\left[x,\, \frac{1}{2}-y,\, z\right]$	`[13,29]`
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[14,30]`
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	`[15,31]`
16	$[y,\, x,\, z]$	`[16,32]`